EXPERIMENT - 21

Set 21: Website Traffic Analysis

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

Question 18

In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

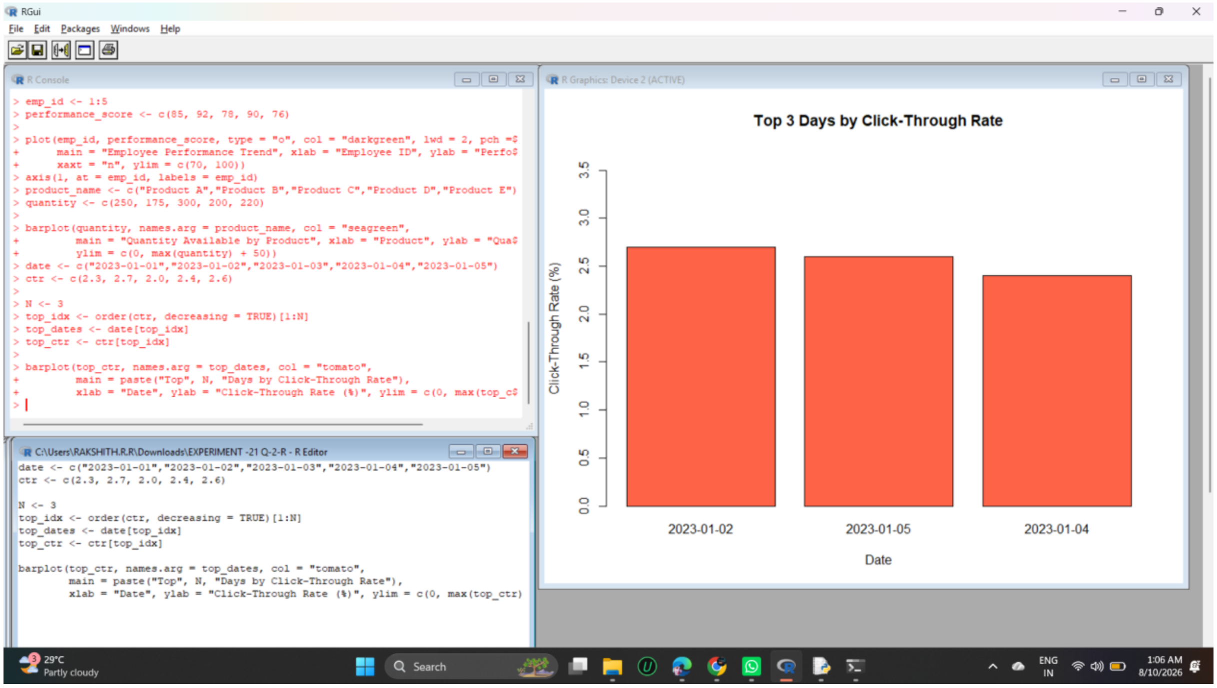
R Code

```
date <- c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

N <- 3
top_idx <- order(ctr, decreasing = TRUE)[1:N]
top_dates <- date[top_idx]
top_ctr <- ctr[top_idx]

barplot(top_ctr, names.arg = top_dates, col = "tomato",
        main = paste("Top", N, "Days by Click-Through Rate"),
        xlab = "Date", ylab = "Click-Through Rate (%)", ylim = c(0, max(top_ctr) + 1))
```

Output



Question 19

In Python, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

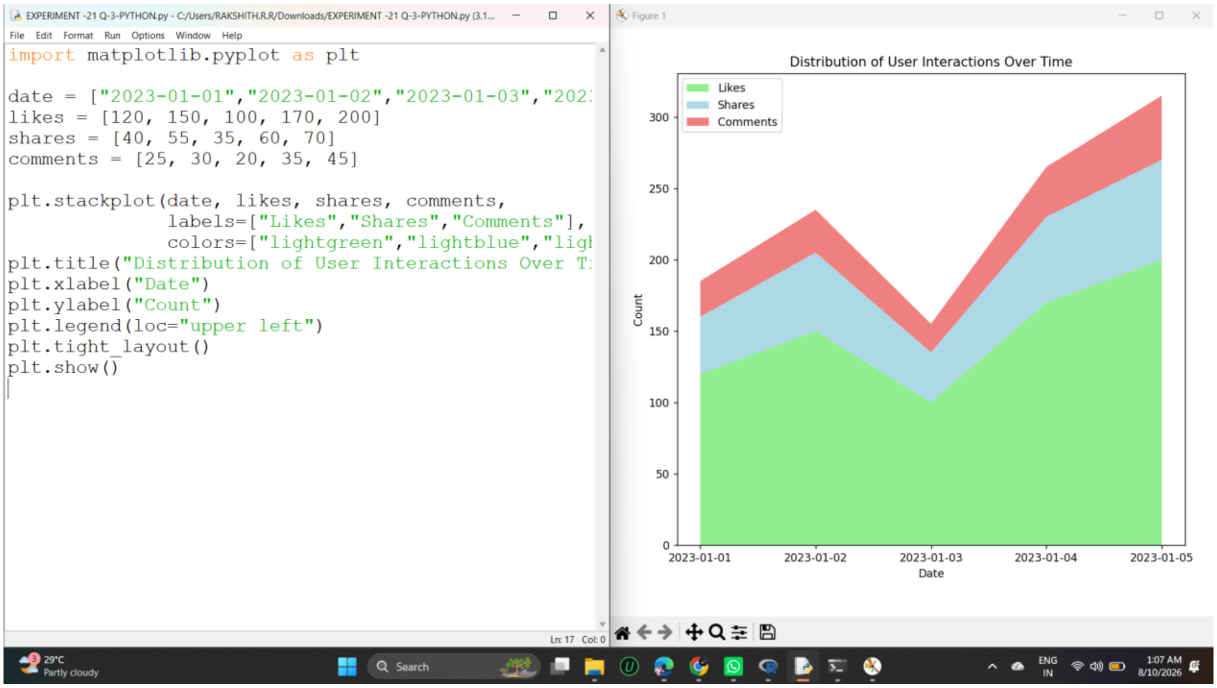
Python Code

```
import matplotlib.pyplot as plt

date = ["2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05"]
likes = [120, 150, 100, 170, 200]
shares = [40, 55, 35, 60, 70]
comments = [25, 30, 20, 35, 45]

plt.stackplot(date, likes, shares, comments,
        labels=["Likes","Shares","Comments"],
        colors=["lightgreen","lightblue","lightcoral"])
plt.title("Distribution of User Interactions Over Time")
plt.xlabel("Date")
plt.ylabel("Count")
plt.legend(loc="upper left")
plt.tight_layout()
plt.show()
```

Output


EXPERIMENT -21 Q-3-PYTHON.py - C:/Users/RAKSHITH.R.R/Downloads/EXPERIMENT -21 Q-3-PYTHON.py (3.1...
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
date = ["2023-01-01","2023-01-02","2023-01-03","202
likes = [120, 150, 100, 170, 200]
shares = [40, 55, 35, 60, 70]
comments = [25, 30, 20, 35, 45]
plt.stackplot(date, likes, shares, comments,
labels=["Likes","Shares","Comments"],
colors=["lightgreen","lightblue","ligl
plt.title("Distribution of User Interactions Over T
plt.xlabel("Date")
plt.ylabel("Count")
plt.legend(loc="upper left")
plt.tight_layout()
plt.show()
Ln: 17 Col: 0
Figure 1
Distribution of User Interactions Over Time
Likes
Shares
Comments
Count
Date
2023-01-01
2023-01-02
2023-01-03
2023-01-04
2023-01-05
29°C
Partly cloudy
Search
ENG
IN
1:07 AM
8/10/2026